

Category-Theoretic Improvements to *Extending Intuitionistic Logic to a Probability Calculus (in Lean 4)* A Sequentially Organized List

This document lists categorical improvements to the anonymous manuscript, organized *sequentially* so that each item depends only on earlier items. Part I fixes terminology, Part II collects structural results, Part III addresses the monad theory, Part IV compares with external frameworks, and Part V sketches speculative directions. Each item states a problem, the suggested addition, and a note on why it belongs at that position in the sequence.

Contents

1	Part I: Foundational Clarifications	2
1.1	Item 1: Frame vs. Locale Terminology	2
1.2	Item 2: Variance of the Valuation Functor	2
1.3	Item 3: Pushforward vs. Pullback Terminology	2
2	Part II: Structural Results	3
2.1	Item 4: The Category of Valuations	3
2.2	Item 5: The Nucleus $\neg\neg$ and the Booleanization	3
2.3	Item 6: Sharp Valuations as Points	3
2.4	Item 6b: Scott Continuity as Scott Topology (Restored)	4
2.5	Item 6c: Sierpinski Space as Classifying Locale (Restored)	4
2.6	Item 7: The Cox Product Rule as a Monoid Embedding	4
2.7	Item 8: Modularity as Bilinearity (Strengthened)	5
3	Part III: The Monad Question	5
3.1	Item 9: The Endofunctor Problem	5
3.2	Item 10: Unit as Natural Transformation	5
3.3	Item 11: Kock’s Theorem and Commutativity	6
3.4	Item 12: Countable-Mixture Fragment and Kleisli Composition (Strengthened)	6
3.5	Item 13: Algebras of the Monad	6
4	Part IV: Comparisons and External Frameworks	6
4.1	Item 14: Comparison with the Giry Monad	6
4.2	Item 15: Classifying Locale of the Geometric Theory of Valuations (Strengthened)	7
4.3	Item 16: Stone Duality and Chaitin’s Ω	7

5 Part V: Speculative and High-Impact Contributions	7
5.1 Item 17: A 2-Categorical Perspective	7
5.2 Item 18: Duality Between Belief and Plausibility	8
Summary Table	8
How to Use This Sequence	8

1 Part I: Foundational Clarifications

These items are self-contained and fix terminology. Every later item depends on them.

1.1 Item 1: Frame vs. Locale Terminology

Problem. The paper uses “frame” throughout but occasionally slips into locale-theoretic language (“points of a locale,” “classifying locales”) without defining the categorical relationship.

Suggested addition.

- Define **Frm** as the category of frames with frame homomorphisms (preserving finite meets and arbitrary joins).
- Define **Loc** = **Frm**^{op}.
- State the adjunction $O \dashv \text{pt}$ between **Top** and **Loc**.
- Note that a *point* of a locale Ω is a locale morphism $\mathbf{1} \rightarrow \Omega$, equivalently a frame homomorphism $\Omega \rightarrow \mathbf{2}$.

Why here. Every later categorical statement depends on knowing which category we work in.

1.2 Item 2: Variance of the Valuation Functor

Problem. The paper says “ $\text{Val}(-)$ is a functor on locales” but does not specify variance or target category.

Suggested addition.

- A frame homomorphism $f : \Omega \rightarrow \Omega'$ induces $f^* : \text{Val}(\Omega') \rightarrow \text{Val}(\Omega)$ by $v \mapsto v \circ f$.
- Hence $\text{Val} : \mathbf{Frm} \rightarrow \mathbf{Set}$ is *covariant*, and $\text{Val} : \mathbf{Loc} \rightarrow \mathbf{Set}^{\text{op}}$ is contravariant.
- State explicitly: “Throughout, Val is a functor $\mathbf{Frm} \rightarrow \mathbf{Set}$.”

Why here. Fixes variance, which all later functoriality and monad claims depend on.

1.3 Item 3: Pushforward vs. Pullback Terminology

Problem. The paper calls $v \mapsto v \circ f$ “pushforward,” conflicting with measure-theoretic usage.

Suggested addition.

- In measure theory, pushforward along $f : X \rightarrow Y$ is $f_*\mu = \mu \circ f^{-1}$.

- In locale theory, a locale morphism $f : \Omega \rightarrow \Omega'$ is a frame homomorphism $\Omega' \rightarrow \Omega$, so $v \mapsto v \circ f$ is the direct analogue of pushforward *in the locale direction*.
- Recommend: call it “pushforward along the locale morphism” and note the variance explicitly.

Why here. Depends on Items 1–2; prevents confusion in Section 8 of the paper.

2 Part II: Structural Results

These are concrete categorical constructions that only require Part I.

2.1 Item 4: The Category of Valuations

Problem. No category is defined whose objects are valuations.

Suggested addition. Define Val-category:

- Objects: pairs (Ω, v) with Ω a frame and v a valuation.
- Morphisms: frame homomorphisms $f : \Omega \rightarrow \Omega'$ such that $v = v' \circ f$.
- Prove basic properties: terminal object, products, etc.
- State the projection functor $\pi : \mathbf{Val} \rightarrow \mathbf{Frm}$.

Why here. Depends only on Items 1–3; provides a home for later monad-theoretic statements.

2.2 Item 5: The Nucleus $\neg\neg$ and the Booleanization

Problem. Section 5 uses the Booleanization (regular elements) without naming its categorical status.

Suggested addition.

- State that $\neg\neg : \Omega \rightarrow \Omega$ is a *nucleus* (a monad on Ω in the 2-category of frames).
- The Booleanization is its category of algebras (the fixed points of $\neg\neg$).
- Theorem 23 (total monotonicity) is a statement about the *Kleisli category* of this nucleus.

Why here. Depends only on Items 1–2; clarifies Section 5 of the paper without invoking the monad theory of Val.

2.3 Item 6: Sharp Valuations as Points

Problem. Theorem 12 (sharp valuations = points) is stated set-theoretically.

Suggested addition.

- Rephrase: sharp valuations on Ω are in natural bijection with locale morphisms $\mathbf{1} \rightarrow \Omega$, i.e. with the point functor $\text{pt} : \mathbf{Loc} \rightarrow \mathbf{Set}$.
- Scott-continuous sharp valuations correspond to completely prime points.

Why here. Depends on Item 1; bridges Section 3 of the paper to locale theory without needing the monad.

2.4 Item 6b: Scott Continuity as Scott Topology (Restored)

Problem. The paper uses Scott continuity (Definition 13, Theorem 14, Examples 18–19, Theorem 20) without naming it as the Scott topology or connecting it to the Scott model of the lambda calculus.

Suggested addition.

- Scott continuity is continuity with respect to the *Scott topology* on the frame Ω .
- This is the standard notion of continuity in *domain theory* and the *Scott model* of the lambda calculus.
- Connect Example 18 (top-indicator) to the failure of Scott continuity as the failure of approximation by finite stages.

Why here. Depends on Items 1 and 6; independent of the monad theory. Connects the paper to domain theory and denotational semantics.

2.5 Item 6c: Sierpinski Space as Classifying Locale (Restored)

Problem. Section 7.3 treats the Sierpinski space point-set theoretically, missing its categorical role.

Suggested addition.

- The Sierpinski space $\mathbb{S}$ is the *classifying locale* of the theory of a single proposition.
- The fact that $\neg \text{halts} = \perp$ is a statement about the *internal logic* of the locale.
- Frame this as: the Sierpinski locale classifies *open propositions*, and the halting event is the generic open proposition.
- The uncomputability of the sharp readout (Theorem 44) is then the statement that the *global sections functor* $\Gamma : \mathbf{Loc} \rightarrow \mathbf{Set}$ does not preserve the relevant structure.

Why here. Depends only on Items 1–2; independent of the monad theory. Reframes the entire computability guard in locale-theoretic terms.

2.6 Item 7: The Cox Product Rule as a Monoid Embedding

Problem. Section 6 builds Aczel’s theorem from scratch but does not state its categorical content.

Suggested addition.

- A scale $(S, F, \leq)$ is a totally ordered commutative semigroup.
- Aczel’s theorem gives a monoid embedding $(S, F) \hookrightarrow (\mathbb{R}_{>0}, \times)$.
- The regradation g is a monoid homomorphism.
- This makes the product-rule half of Cox’s theorem a statement about ordered semigroups, independent of Ω .

Why here. Depends only on basic algebra; independent of the monad theory. Clarifies Section 6 of the paper.

2.7 Item 8: Modularity as Bilinearity (Strengthened)

Problem. Modularity is presented as an axiom but its categorical content is not named.

Suggested addition.

- The frame Ω has two monoidal structures: $(\Omega, \sqcap, \top)$ and $(\Omega, \sqcup, \perp)$.
- Modularity says v is a *monoid homomorphism* for both, up to normalization.
- In monoidal-category language: Ω is a lattice with two monoidal structures, and v is a bilinear map.
- The failure of the sum rule is the failure of $\sqcup$ and $\sqcap$ to distribute (i.e. Ω is not a Boolean algebra).

Why here. Depends on Item 6; clarifies Sections 2 and 6 of the paper. The monoidal-category language is essential for a category theorist.

3 Part III: The Monad Question

These are the core categorical contributions, built on Parts I–II.

3.1 Item 9: The Endofunctor Problem

Problem. A monad requires an endofunctor $T : \mathcal{C} \rightarrow \mathcal{C}$. The paper has $\text{Val} : \mathbf{Frm} \rightarrow \mathbf{Set}$, which is not an endofunctor.

Suggested addition.

- Either (a) construct a locale structure on $\text{Val}(\Omega)$ (the classifying locale of the geometric theory of valuations), making $\text{Val} : \mathbf{Loc} \rightarrow \mathbf{Loc}$; or
- (b) downgrade the claim to “candidate monad data” and state the monad laws as conjectural.

Why here. Depends on Items 1–4; first genuinely monad-theoretic question.

3.2 Item 10: Unit as Natural Transformation

Problem. Point valuations δ_p require points, which a general locale need not have.

Suggested addition.

- Either restrict to spatial locales and prove naturality there;
- Or use the classifying-locale construction to define $\eta : \text{id} \Rightarrow \text{Val}$ point-freely.

Why here. Depends on Item 9; the unit problem is only meaningful once the endofunctor is fixed.

3.3 Item 11: Kock’s Theorem and Commutativity

Problem. Kock’s correspondence is invoked informally.

Suggested addition.

- Specify the monoidal structure on **Loc** (product of locales = coproduct of frames).
- State Kock’s theorem precisely: a monad is commutative iff it has a Fubini map.
- Theorem 47 (product non-uniqueness) shows $\mathbf{Val}$ is not commutative.

Why here. Depends on Items 9–10; needs the endofunctor and unit in place.

3.4 Item 12: Countable-Mixture Fragment and Kleisli Composition (Strengthened)

Problem. Proposition 48 is informal, and the Kleisli category is never named.

Suggested addition.

- Define the Kleisli category $\mathbf{Kl}(\mathbf{Val}_{\text{count}})$.
- The Kleisli category $\mathbf{Kl}(\mathbf{Val}_{\text{count}})$ is the category of *countably-presented Markov kernels*.
- Prove the four items of Proposition 48 are exactly the monad laws (unit, associativity, commutativity) in this Kleisli category.
- Discuss the quotient by presentation and whether it descends to valuations.

Why here. Depends on Items 9–11; constructive part of the monad theory. The Markov-kernel identification makes the categorical content immediately recognizable.

3.5 Item 13: Algebras of the Monad

Problem. No discussion of Eilenberg–Moore algebras.

Suggested addition.

- Conjecture: finite valuations have convex sets as algebras; countably-presented valuations have countable convex sets; general valuations have Dempster–Shafer belief functions or interval-valued expectations.
- Connect to Section 5 of the paper (DS reading).

Why here. Depends on Items 9–12; algebras are meaningless without the monad.

4 Part IV: Comparisons and External Frameworks

4.1 Item 14: Comparison with the Giry Monad

Problem. The Giry monad lives on **Meas**, not **Loc**.

Suggested addition.

- Either restrict to spatial locales and compare there;
- Or invoke Fritz’s synthetic framework (2020) explicitly and say $\mathbf{Val}$ is a partial analogue.

Why here. Depends on Items 9–13; needs the monad theory in place first.

4.2 Item 15: Classifying Locale of the Geometric Theory of Valuations (Strengthened)

Problem. Mentioned in the Conclusion of the paper but never developed.

Suggested addition.

- State the geometric theory explicitly (sorts, axioms for monotonicity, normalization, modularity).
- Prove that its classifying locale is $\text{Val}(\Omega)$ for Scott-continuous valuations.
- Discuss whether the classifying-locale construction gives a *2-monad* on **Loc**, i.e. a 2-monad on the 2-category of geometric theories.

Five justifications for high impact.

1. Gives a precise categorical home for Val .
2. Connects to the existing literature on classifying locales (Coquand–Spitters, Vickers).
3. Makes the monad question rigorous.
4. Provides a point-free analogue of the Giry monad.
5. Explains why the diffuse part (Example 18) obstructs integration (the multiplication of the monad).

Why here. Depends on Items 9–14; this is the high-impact categorical contribution.

4.3 Item 16: Stone Duality and Chaitin’s Ω

Problem. Section 7.3 is point-set.

Suggested addition.

- Cantor space is the Stone space of the free Boolean algebra on countably many generators.
- Chaitin’s Ω is a valuation on its clopens.
- The uncomputability of Ω is a statement about the non-effectiveness of the Stone duality.

Why here. Depends on Items 1–6; can be done independently of the monad theory but is best placed after to avoid disrupting the flow.

5 Part V: Speculative and High-Impact Contributions

5.1 Item 17: A 2-Categorical Perspective

Suggested addition.

- The classifying-locale construction is a 2-functor from geometric theories to locales.
- The monad structure on Val would be a 2-monad on the 2-category of geometric theories.
- Speculate on the 2-categorical structure of valuations.

Why last. Depends on Item 15; highly speculative.

5.2 Item 18: Duality Between Belief and Plausibility

Suggested addition.

- The interval $[v(a), 1 - v(\neg a)]$ might be a comonadic structure dual to the monad of valuations.
- Speculate on a categorical duality between belief and plausibility.

Why last. Depends on Items 5 and 13; speculative.

Summary Table

Order	Topic	Depends on
1	Frame vs. locale terminology	—
2	Variance of Val	1
3	Pushforward vs. pullback	1, 2
4	Category of valuations	1–3
5	Nucleus $\neg\neg$ and Booleanization	1–2
6	Sharp valuations as points	1
6b	Scott continuity as Scott topology	1, 6
6c	Sierpinski as classifying locale	1–2
7	Cox product rule as monoid embedding	—
8	Modularity as bilinearity	6
9	The endofunctor problem	1–4
10	Unit as natural transformation	9
11	Kock’s theorem and commutativity	9–10
12	Countable-mixture fragment and Kleisli	9–11
13	Algebras of the monad	9–12
14	Comparison with Giry monad	9–13
15	Classifying locale of valuations	9–14
16	Stone duality and Chaitin’s Ω	1–6
17	2-categorical perspective	15
18	Belief/plausibility duality	5, 13

How to Use This Sequence

- **Items 1–8** are self-contained clarifications and structural results. They can be addressed immediately without committing to any monad theory.
- **Items 6b, 6c** are restored from the initial review; they reframe Scott continuity and the computability guard in locale-theoretic terms.
- **Items 9–13** form the core monad-theoretic contribution. They require Items 1–8 as prerequisites.

- **Items 14–16** are comparisons and external frameworks. They require the monad theory to be meaningful.
- **Items 17–18** are speculative and can be included as “future work” or omitted.

A single new section integrating all of this could be structured as:

Section 8 (revised): Categorical Structure of Valuations

- 8.1 Frames, locales, and variance (Items 1–3)
- 8.2 The category of valuations (Item 4)
- 8.3 The nucleus $\neg\neg$ and the Booleanization (Item 5)
- 8.4 Sharp valuations and Scott continuity (Items 6, 6b)
- 8.5 The Sierpinski locale and computability (Item 6c)
- 8.6 The endofunctor and the monad question (Items 9–10)
- 8.7 Kock’s theorem and the failure of Fubini (Item 11)
- 8.8 The countable-mixture fragment (Item 12)
- 8.9 Algebras and the DS reading (Item 13)
- 8.10 Comparison with the Giry monad (Item 14)
- 8.11 The classifying locale (Item 15)